

Phase 4: Project Planning Phase

Project title :: Script-Controlled ACL - Restrict Record Access Based on Field Value

College Name: Government Arts and Science College, Manur.

College Code: MSU142

Department: B.SC Computer Science

Team ID: SWTID-2026-9029

TeamLeader: Subbulakshmi M NM ID: 727F7AC2ABD1D8C8DF6496517D478773

Team member: Arumuga Kani S NM id: 601EE17CA3AAD3AB4EA4C8F841A16DD1

Team member: Mariammal C NM id: BBE8324EB82C757A1054DE3C9951281A

Team member: Esakki Ganesh P NM id: A705CB2C71F3AF38AF2473EFB5AE7725

Team member: Akalya R NM id: B3F460E08A938D39C7F40868EA8EC1C6

The Project Planning Phase is the stage where the complete development process of the Script-Controlled ACL – Restrict Record Access Based on Field Value project is planned before implementation. It provides a clear roadmap for developing, testing, documenting, and demonstrating the system.

The main purpose of this phase is to identify what needs to be developed, what resources are required, how the work will be carried out, and how the project will achieve its security objectives.

4.1 Project Objective Planning

The primary objective is to develop a system that restricts access to individual records according to specific field values. The system should provide controlled and secure access to authorized users.

4.2 Requirement Planning

All necessary requirements are identified before development. These include:

- User requirements
- Functional requirements
- Security requirements
- Database requirements
- Software requirements
- Testing requirements

4.3 Resource Planning

The resources required for project development are identified. This includes the development environment, database, scripting environment, computer system, and other required software tools.

4.4 User and Role Planning

Different users and their access levels are planned. The system determines which users should be permitted to access particular records.

4.5 Field Selection Planning

A suitable field is selected for controlling record access. The field value is used by the ACL script to determine whether the access request should be accepted or rejected.

4.6 Access Rule Planning

The access conditions are defined before writing the ACL script. These rules specify which field values should allow access and which values should restrict access.

4.7 System Workflow Planning

The complete workflow is planned from the user's request to the final access decision.

User Login → Request Record → Identify Field Value → Execute ACL → Validate Condition → Grant / Deny Access

4.8 Development Task Planning

The development work is divided into smaller tasks to make the project easier to manage:

1. Configure the system.
2. Create records and fields.
3. Configure users.
4. Define access conditions.
5. Develop the ACL script.
6. Apply the ACL.

7. Test the system.
8. Correct errors.
9. Prepare documentation.

4.9 Testing Plan

Testing activities are planned before implementation is completed. Different combinations of users, records, and field values are considered to verify both permitted and restricted access.

4.10 Security Planning

Security is considered throughout the project. The planning process ensures that unauthorized users cannot access records simply by requesting them directly.

4.11 Risk Planning

Possible risks such as incorrect configuration, invalid field values, script errors, and incorrect access decisions are identified. Methods for detecting and correcting these issues are planned.

4.12 Time Schedule Planning

A suitable schedule is prepared for completing each project phase. Time is allocated for analysis, design, development, testing, documentation, and demonstration.

4.13 Documentation Planning

The required project report sections are planned in advance. These include:

- Introduction
- Objectives

- Requirements
- Design
- Development
- Testing
- Results
- Screenshots
- Advantages
- Conclusion

4.14 Final Review Planning

Before final submission, the completed system is planned to undergo a final review. The ACL rules, test results, documentation, and demonstration are checked for completeness.

Overall Planning Flow

Project Objectives → Requirements → Resources → Users & Roles → Field Selection → Access Rules → Development Plan → Testing Plan → Documentation → Final Review

Expected Outcome

The Project Planning Phase provides a structured roadmap for the entire project. It ensures that the Script-Controlled ACL is developed systematically, tested properly, documented clearly, and prepared for final demonstration.